

Group Exercise: Predicting Product Sales Using Linear Regression

Topic

Linear Regression: Univariate vs Multivariate Regression

Dataset

oil_sales_assignment_dataset.csv

Learning Objectives

By the end of this exercise, your group should be able to:

- Explore a real-world sales dataset.
 - Build a Univariate Linear Regression model.
 - Build a Multivariate Linear Regression model.
 - Compare model performance.
 - Interpret regression coefficients.
 - Present findings to the class.
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Dataset Description

The dataset contains sales information for different products sold across multiple cities and stores.

Available Columns

Column	Description
city	City where the product was sold
store_name	Store name

manufacturer	Product manufacturer
brand	Product brand
class	Product category
size	Package size
sku	Product identifier
price_bracket	Price category
year	Sales year
month	Sales month
value_sales	Total sales value (Target Variable)
volume_sales	Number of units sold
average_price	Average selling price

Scenario

Your company wants to predict **Value Sales** for future products.

You have been hired as a Data Scientist to determine which factors influence sales.

Part 1: Data Exploration (10 Marks)

Task 1

Load the dataset.

Display

- first five rows
 - last five rows
-

Task 2

Answer the following:

1. How many rows are there?
 2. How many columns?
 3. Which columns are numerical?
 4. Which columns are categorical?
-

Task 3

Check for missing values.

Are there any missing values?

Task 4

Generate descriptive statistics.

Hint

```
describe()
```

Part 2: Data Preparation

Since Linear Regression requires numerical data,

convert all categorical variables into numerical values using LabelEncoder.

Encode the following columns:

- city
- store_name
- manufacturer
- brand
- class

- size
 - sku
 - price_bracket
-

Part 3: Univariate Linear Regression

Objective

Predict

```
Value Sales
```

using only

```
Average Price
```

Task 1

Create

```
X = average_price  
  
y = value_sales
```

Task 2

Split the dataset into

- 80% Training
- 20% Testing

Use

```
random_state = 42
```

Task 3

Train a Linear Regression model.

Task 4

Write the regression equation.

Example

```
Sales = b0 + b1 × Average Price
```

Task 5

Interpret the coefficient.

Explain in one sentence:

What happens to sales when the average price increases by one unit?

Task 6

Evaluate the model using

- RMSE
 - R^2 Score
-

Task 7

Draw

- Scatter Plot
 - Regression Line
-

Part 4: Multivariate Linear Regression

Objective

Predict

Value Sales

using multiple variables.

Suggested features

```
average_price  
volume_sales  
year  
month  
city_encoded  
manufacturer_encoded  
brand_encoded  
class_encoded  
price_bracket_encoded
```

(Your group may include additional encoded variables if appropriate.)

Task 1

Prepare the feature matrix.

Task 2

Split the dataset.

Task 3

Standardize the features using

Task 4

Train the model.

Task 5

Display the intercept.

Task 6

Display all coefficients in a DataFrame.

Sort them from largest to smallest based on the absolute coefficient value.

Task 7

Evaluate using

- RMSE
- R^2 Score

Part 5: Model Comparison

Create a comparison table.

Model	RMSE	R^2
Univariate		
Multivariate		

Create two bar charts.

Chart 1

Compare RMSE.

Chart 2

Compare R^2 .

Part 6: Interpretation Questions

Discuss the following questions with your group.

Question 1

Which model performed better?

Why?

Question 2

Why is the Multivariate model expected to perform better than the Univariate model?

Question 3

Which feature appears to have the greatest influence on sales?

Support your answer using the coefficient table.

Question 4

What does a negative coefficient mean?

Question 5

If the R^2 score is 0.82,

what does this tell us about the model?

Question 6

Why should we evaluate the model using the test set instead of the training set?

Deliverables

Each group should submit:

- ✓ Python Notebook (.ipynb)
- ✓ Source Code
- ✓ Comparison Table
- ✓ Regression Plots
- ✓ RMSE and R^2 Results
- ✓ A 5-minute presentation explaining:
 - Dataset
 - Features used
 - Model results
 - Interpretation
 - Final conclusion